BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, KOHAT

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: KT-XI-23-I

Retyped faithfully from the original board paper — same questions, no changes.

Section "A"

Time Allowed: 20 Minutes | **Maximum Marks:** 18 | **Paper Code:** Code: A

Note: Q.1 Attempt all the questions of Section A by filling the correct option i.e. A, B, C or D in the OMR RESPONSE SHEET. No marks will be awarded for cutting, erasing or over writing.

Qi. The metric prefix for 0.000001 is .....

(A) hecto
(B) micro
(C) deca
(D) nano

Qii. $A \cdot (A \times B) = ?$

(A) 0
(B) 1
(C) AB
(D) A^2B

Qiii. To improve the jumping record the long jumper should jump at an angle of .....

(A) 30°
(B) 45°
(C) 60°
(D) 90°

Qiv. When the speed of your car is doubled, by what factor does its kinetic energy increases?.

(A) √2
(B) 2
(C) 4
(D) 8

Qv. The angular speed in radians/hours for daily rotation of our earth is .....

(A) 2π
(B) 4π
(C) π/6
(D) π/12

Qvi. When the radius of the artery is reduced, the blood pressure .....

(A) increased
(B) decreased
(C) remains the same
(D) is zero

Qvii. Tuning of a radio set is an example of .....

(A) mechanical resonance
(B) free vibration
(C) electrical resonance
(D) musical resonance

Qviii. The fundamental frequency of a closed organ pipe is f. If both the ends are opened then its fundamental frequency will be .....

(A) f
(B) 0.5 f
(C) 2f
(D) 4f

Qix. The tip of a needle does not give a sharp image. It is due to .....

(A) polarization
(B) interference
(C) diffraction
(D) refraction

Qx. Maximum work can be obtained in the process called .....

(A) cyclic
(B) Isothermal
(C) Adiabatic
(D) isochoric

Qxi. The rotational analogue of force is .....

(A) work
(B) impulse
(C) torque
(D) momentum

Qxii. $[M^0L^0T^0]$ are dimensions of .....

(A) strain
(B) refractive index
(C) magnification
(D) all of these

Qxiii. The magnitude of a vector, A = 2i + 2j + 2k is .....

(A) $\sqrt[3]{3}$
(B) $\sqrt[3]{2}$
(C) $\sqrt{2}$
(D) $\sqrt{3}$

Qxiv. SI unit of impulse is .....

(A) $kgms^{-2}$
(B) Ns
(C) Ns^{-1}
(D) Nm

Qxv. Angular velocity has the same dimensions as that of .....

(A) velocity
(B) frequency
(C) time period
(D) amplitude

Qxvi. According to equation of continuity, Av = constant. This constant is equal to

(A) volume of fluid
(B) mass of fluid
(C) density of fluid
(D) volume flow rate

Qxvii. The time period of a simple pendulum is 2 seconds. If its length is increased by four times, then its period become …….

(A) 16 s
(B) 12s
(C) 8s
(D) 4s

Qxviii. Triple point of water is

(A) 273.16 °C
(B) 372.16 k
(C) 273.16°F
(D) 273.16k